

Paper II - Urology (MCh Urology, Summer 2015)

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II-1. Describe medical management of BPH and role of various surgical modalities in its management.

Type: LAQ (25 marks)

Introduction / Definition

- **BPH (benign prostatic hyperplasia)** is a histological diagnosis: stromal and epithelial hyperplasia of the transition zone.
- Enlargement can cause **BPO (benign prostatic obstruction)** and **LUTS (lower urinary tract symptoms)**.
- Goals: relieve bother and prevent progression to AUR (acute urinary retention) or surgery.

Pathophysiology relevant to therapy

- **Static component** (gland bulk, DHT (dihydrotestosterone)-dependent): 5-ARIs (5 α -reductase inhibitors) and surgery.
- **Dynamic component** (α 1A-adrenergic smooth-muscle tone): α -blockers.
- **Bladder response** (detrusor overactivity): antimuscarinics, β 3-agonists.

Evaluation

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, FVC (frequency-volume chart), Qmax (maximum flow rate) and PVR (post-void residual). PVR >300 mL is managed as chronic retention.
- AUA (American Urological Association): measure prostate volume before surgery; it decides the modality.

Medical management

1. **Watchful waiting:** mild or non-bothersome symptoms; lifestyle advice.
2. **α 1-blockers:** first line for bothersome LUTS; act within days.

Drug	Dose	Note
Tamsulosin	0.4 mg daily	Highest IFIS risk
Alfuzosin	10 mg daily	Least ejaculatory dysfunction
Silodosin	8 mg daily	Most α 1A selective; most anejaculation

- They do **not** shrink the prostate or prevent AUR or surgery.
- **IFIS (intraoperative floppy iris syndrome)**: stopping the drug before cataract surgery does not remove the risk, so defer starting it if cataract surgery is planned.

3. 5-ARIs: finasteride 5 mg or dutasteride 0.5 mg.

- For enlarged glands (AUA: >30 mL or PSA >1.5 ng/mL; EAU (European Association of Urology): >40 mL).
- Shrink the prostate about 18–25% (maximal at 6 months) and halve PSA.
- **PLESS (Proscar Long-Term Efficacy and Safety Study)**: finasteride reduced AUR or surgery by 51%.
- Side effects: low libido, ED (erectile dysfunction), ejaculatory dysfunction, gynaecomastia.

4. Combination therapy for men at high risk of progression (>30–40 mL, PSA >1.5 ng/mL)

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell 39% with doxazosin, 34% with finasteride and **66% with combination**. Only finasteride or combination reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.

5. Storage symptoms: add an antimuscarinic or mirabegron; trials included only men with low PVR (<150–200 mL).

6. Tadalafil 5 mg daily: improves LUTS with or without ED; not with nitrates.

7. Nocturnal polyuria (night urine >33% of 24-h output): low-dose desmopressin; monitor sodium, especially if ≥65 years.

LUTS → IPSS, DRE, urinalysis, PSA, PVR
 Mild → watchful waiting
 Voiding → α -blocker (+ 5-ARI if >30–40 mL)
 Storage → add antimuscarinic or β 3-agonist
 Failure or complications → surgery by prostate size

Surgical management

Indications: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, and LUTS failing medical therapy.

Modality	Size (AUA 2023)	Key feature
TURP (transurethral resection of prostate)	30–80 mL	Reference standard; retrograde ejaculation ~two-thirds
TUIP (transurethral incision of prostate)	≤30 mL	Less retrograde ejaculation; more reoperation
PVP (photoselective vaporization)	≤80 mL	Safe on anticoagulants; no histology
HoLEP (holmium laser enucleation)	Any size	Lowest retreatment, least bleeding; learning curve
Simple prostatectomy	>80 mL	Complete removal; more morbidity if open
Aquablation	30–80 mL (consider 80–150)	Heat-free; preserves ejaculation; bleeding
PUL (prostatic urethral lift), Rezūm (water vapour)	30–80 mL (PUL: no median lobe)	Preserve sexual function; higher retreatment

- **TUR syndrome** (monopolar TURP): hypo-osmolar irrigant (glycine, water) is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk

rises with gland >45 mL and resection >90 minutes. Treat by stopping surgery, diuretics, and 3% saline if severe, correcting slowly.

- **Bipolar TURP** uses saline, eliminating TUR syndrome, with less bleeding.

Landmark trials

- **GOLIATH**: 180 W PVP non-inferior to TURP, with shorter stay.
- **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior to TURP, with less retrograde ejaculation (10% vs 36%).

Complications of surgery

- Early: bleeding, TUR syndrome. Late: stricture, bladder neck contracture, retreatment.

Follow-up

- Reassess at 4–12 weeks on α -blockers and 3–6 months on 5-ARIs, then yearly.

Recent advances

- **Optilume BPH** (paclitaxel-coated balloon), approved 2023; PAE (prostate artery embolisation).

Diagram to draw: Size-based algorithm: ≤ 30 mL TUIP $\rightarrow$ 30–80 mL TURP/PVP/MIST (minimally invasive surgical therapy) $\rightarrow$ >80 mL enucleation or simple prostatectomy.

Key points to remember:

- IPSS 0–35: 0–7 mild, 8–19 moderate, 20–35 severe.
 - 5-ARIs shrink the prostate ~20%, halve PSA; PLESS cut AUR/surgery 51%; MTOPS combination cut progression 66%.
 - AUA sizes: TUIP ≤ 30 ; TURP/PVP/PUL/Rezūm/Aquablation 30–80; HoLEP any size; simple prostatectomy >80 mL.
 - TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.
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II-2. Discuss metabolic evaluation in urinary calculus disease.

Type: LAQ (25 marks)

Introduction

- Urolithiasis is a chronic systemic metabolic disease; up to 50% recur within 5–10 years without prevention.

Physicochemical basis

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention (often on **Randall plaque**).
- **Metastable zone** (between solubility and formation products): no spontaneous nucleation, but existing crystals grow. Above the formation product, nucleation is spontaneous.
- **Inhibitors** (citrate, magnesium, Tamm-Horsfall protein) raise the supersaturation needed for crystallisation.

Who needs evaluation

- **Screening evaluation**: every first-time stone former (AUA (American Urological Association), EAU (European Association of Urology)).
- **24-h urine evaluation**: high-risk and recurrent formers.
- **High risk**: children, family history, recurrent stones, solitary kidney; uric acid, cystine or infection stones; GI (gastrointestinal) disease or bypass, hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, metabolic syndrome.

Screening evaluation (all patients)

- **History**: diet, bowel disease, lithogenic drugs (calcium, vitamins C and D, acetazolamide, steroids).
- **NCCT (non-contrast computed tomography)**: uric acid stones are usually <500 HU (Hounsfield units).
- **Serum**: creatinine, electrolytes, calcium, uric acid, bicarbonate; PTH (parathyroid hormone) if calcium is high-normal or raised.
- **Urine**: pH, crystals (hexagonal = cystine), culture.
- **Stone analysis** (infrared spectroscopy or X-ray diffraction): uric acid suggests a metabolic cause, cystine a genetic one, struvite infection.

Comprehensive evaluation: 24-h urine

- One or two collections on usual diet, ≥ 4 weeks after the stone event, when stone-free and infection-free.
- Check completeness by urinary creatinine (men ~ 20 –25, women ~ 15 –20 mg/kg/day).

Parameter	Desirable (per day)
Volume	>2.5 L (low volume is commonest)
Calcium	<250 mg (women), <300 mg (men), or <4 mg/kg
Oxalate	<40 mg
Citrate	>320 mg
Uric acid	<600 mg (labs: <750 women, <800 men)
Sodium	<100 mmol (salt raises calcium excretion)
pH	5.8–6.2 (<5.5 uric acid; >6.5–7 calcium phosphate, RTA, infection)
Cystine	Negative screen

Interpretation

1. **Hypercalciuria** (commonest): Pak's absorptive (suppressed PTH), renal leak (high PTH, normal serum calcium) and resorptive (primary hyperparathyroidism: high calcium and PTH) types.
2. **Hyperoxaluria** (>40 mg/day): dietary; **enteric** (fat malabsorption or gastric bypass: fatty acids bind calcium, leaving oxalate free for absorption; often 60–70 mg/day, with low citrate and volume); **primary** (hepatic overproduction, often >100 mg/day, childhood onset).
3. **Hypocitraturia** (<320 mg/day): idiopathic, distal RTA, chronic diarrhoea, hypokalaemia, high animal protein.
4. **Hyperuricosuria** promotes calcium oxalate stones. **Uric acid stones** are driven mainly by urine pH <5.5 (metabolic syndrome, diabetes, gout).
5. **Distal RTA**: urine pH >6, hypocitraturia, calcium phosphate stones; in incomplete forms an ammonium chloride load fails to lower pH below 5.5.
6. **Cystinuria**: autosomal recessive transport defect of cystine, ornithine, lysine, arginine. Solubility is about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
7. **Infection stones**: urease-producing organisms, pH >7, struvite.

Stone episode → screening evaluation

Low risk → general advice (urine >2.5 L, less salt and animal protein, normal calcium)

High risk → 24-h urine → specific abnormality → targeted therapy

→ repeat 24-h urine within 6 months → yearly

Treatment linked to abnormality

Abnormality	Treatment
Hypercalciuria	Sodium ≤2.3 g/day, normal calcium; thiazide (HCTZ (hydrochlorothiazide) 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate
Primary hyperparathyroidism	Parathyroidectomy
Hypocitraturia, distal RTA	Potassium citrate 30–60 mEq/day
Enteric hyperoxaluria	Low fat and oxalate; calcium with meals ; magnesium; liquid potassium citrate
Primary hyperoxaluria type 1	Pyridoxine, lumasiran ; liver ± kidney transplant
Hyperuricosuria	Low purine; allopurinol 300 mg/day
Uric acid stones	Potassium citrate to pH 6.0–6.5
Cystinuria	Urine ≥3 L/day, low sodium; alkalinise to pH 7.0–7.5 ; tiopronin if refractory
Infection stones	Complete clearance, culture-specific antibiotics; AHA (acetohydroxamic acid) if surgery impossible

Landmark trials

- **Borghi 1996**: urine >2 L reduced 5-year recurrence (12% vs 27%).
- **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low-calcium diet.

Special groups

- **Children**: all need full evaluation (spot ratios if not toilet-trained).
- **Pregnancy**: defer 24-h urine until after delivery (physiological hypercalciuria).

Follow-up

- 24-h urine within 6 months of therapy, then yearly.

Recent advances

- **RNAi (ribonucleic acid interference)** drugs (lumasiran, nedosiran) for primary hyperoxaluria.
- **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence versus placebo.

Diagram to draw: Supersaturation zones; evaluation algorithm.

Key points to remember:

- 24-h urine on usual diet, ≥ 4 weeks after event; check creatinine for adequacy.
 - Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, Ca $<250/300$ mg (or <4 mg/kg).
 - Distal RTA: NH_4Cl load fails to lower pH <5.5 ; cystinuria: cystine <250 mg/L, pH 7.0–7.5, tiopronin.
 - Borghi 1996: fluid halves recurrence; repeat 24-h urine within 6 months.
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II-3a. Management of priapism

Type: SAQ (10 marks)

Definition

- A full or partial erection lasting **≥4 hours** beyond, or unrelated to, sexual stimulation (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America)).

Classification

Feature	Ischaemic (low-flow)	Non-ischaemic (high-flow)	Stuttering
Mechanism	Veno-occlusive compartment syndrome	Cavernous artery-sinusoidal fistula	Recurrent ischaemic episodes
Cause	ICI (intracavernosal injection), SCD (sickle cell disease), antipsychotics	Perineal trauma	SCD
Penis	Rigid, painful	Partially rigid, painless	Painful episodes
Urgency	Emergency	Elective	Treat episodes as ischaemic

Diagnosis

- Corporal blood gas** (key test): ischaemic is dark, pO₂ <30 mmHg, pCO₂ >60 mmHg, pH <7.25; non-ischaemic resembles arterial blood.
- Colour Doppler** if equivocal: absent flow (ischaemic) or high flow/fistula (non-ischaemic).

Ischaemic priapism

Penile block → aspiration ± saline irrigation
→ Phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h
→ Failure → distal shunt ± tunnelling → proximal shunt
→ >36 h or refractory → consider immediate prosthesis

- Aspiration alone succeeds in under a quarter; add phenylephrine and monitor blood pressure and pulse.
- Distal shunts:** Winter, Ebbehøj, T-shunt, Al-Ghorab; corporal **tunnelling** improves success. **Proximal:** Quackels, Grayhack.
- Duration predicts ED (erectile dysfunction):** 0% ≤12 h, 22% at 12–24 h, 56% at 24–36 h, 100% beyond 36 h.
- Early prosthesis** causes less shortening and fewer complications than delayed insertion into fibrotic corpora.

Non-ischaemic priapism

- Observe; about 60% resolve. If persistent at 4 weeks, **selective embolisation**; repeat embolisation rather than surgery for recurrence.

Stuttering priapism

- Prevent with daily PDE5i (phosphodiesterase-5 inhibitor, which resets PDE5 regulation), pseudoephedrine or self-injected phenylephrine.

Diagram to draw: Penile cross-section showing distal vs proximal shunts.

Key points to remember:

- Ischaemic: $pO_2 < 30$, $pCO_2 > 60$, $pH < 7.25$ - emergency.
 - Phenylephrine 100–500 $\mu\text{g/mL}$, 1 mL q3–5 min up to 1 h.
 - Distal shunt $\pm$ tunnelling before proximal; prosthesis if > 36 h.
 - High-flow: observe ($\sim 60\%$ resolve), then embolise.
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II-3b. Renal tubular acidosis

Type: SAQ (10 marks)

Definition

- **RTA (renal tubular acidosis)** is a hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular hydrogen-ion secretion or bicarbonate reabsorption.

Comparison of types

Feature	Type 1 (distal)	Type 2 (proximal)	Type 4 (hyperkalaemic)
Defect	H ⁺ secretion by α -intercalated cell	Proximal HCO ₃ reabsorption	Aldosterone deficiency or resistance
Urine pH	Always >6; never <5.5	<5.5 at steady state	Low
Serum potassium	Low	Low	High
Urine citrate	Profoundly low	Near normal	–
Stones	Up to 70%: calcium phosphate, nephrocalcinosis	Uncommon	Uncommon
Treatment	Potassium citrate	High-dose alkali plus potassium	Fludrocortisone, low-potassium diet

- **Causes:** distal – Sjögren, lupus, inherited; proximal – Fanconi syndrome; type 4 – diabetic nephropathy, obstruction.
- **Why distal RTA forms stones:** fixed alkaline urine, profound hypocitraturia (acidosis increases tubular citrate reabsorption) and hypercalciuria from bone buffering.
- **Incomplete distal RTA:** normal serum bicarbonate, but urine cannot be acidified below 5.5 after an acid load.

Diagnosis

- **UAG (urine anion gap) = (Na + K) - Cl**, reflects urinary ammonium. Negative: GI (gastrointestinal) loss. Positive: renal cause (distal or type 4 RTA).
- **Ammonium chloride load:** failure to lower urine pH below 5.5 confirms distal RTA.

Urological relevance

- Obstructive uropathy causes hyperkalaemic distal RTA, which improves after relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).
- Potassium citrate is preferred to sodium alkali, because sodium raises calcium excretion.

Recent advances

- Genetic testing for inherited distal RTA.

Diagram to draw: Nephron showing defect sites: proximal tubule (type 2), α -intercalated cell (type 1), aldosterone action (type 4).

Key points to remember:

- Type 1: urine pH >6, hypokalaemia, hypocitraturia, calcium phosphate stones (up to 70%).
- Type 4: hyperkalaemia, low urine pH; seen in obstructive uropathy.
- Positive UAG = renal cause; NH₄Cl load – pH fails to fall <5.5 in distal RTA.

II-3c. Millin's prostatectomy

Type: SAQ (10 marks)

Definition / History

- A **retropubic extravesical simple prostatectomy**: the adenoma is enucleated through a transverse anterior capsular incision, without opening the bladder (Terence Millin, 1945).
- **Freyer's** suprapubic operation removes the adenoma through an opened bladder.

Indications

- Bothering LUTS (lower urinary tract symptoms) or complications of obstruction with a gland **>80-100 g**; or when lithotomy is impossible.
- Choose suprapubic for a large median lobe, diverticulum or stones. Exclude cancer first.

Steps

1. Lower midline or Pfannenstiel extraperitoneal incision into the **space of Retzius**.
2. Control the dorsal venous complex.
3. **Transverse capsulotomy ~2 cm distal to the bladder neck** between stay sutures. It is transverse because a longitudinal incision may extend into the sphincter.
4. Enter the plane between adenoma and surgical capsule; **finger enucleation**.
5. **Divide the urethra at the apex under vision**, protecting the sphincter.
6. Excise the posterior bladder neck lip if prominent and advance its mucosa into the fossa.
7. **Figure-of-eight sutures at 5 and 7 o'clock**, where the prostatic arteries enter.
8. 22 Fr three-way catheter, **watertight capsule closure**, drain.

Millin versus Freyer

Aspect	Millin	Freyer
Bladder	Not opened	Opened
Apex	Divided under vision	Pinched off by finger
Stones, diverticulum, median lobe	Difficult	Ideal
Specific risk	Dorsal venous bleeding	Cystotomy leak

Complications

- Transfusion (<5% with modern technique), urinary leak, infection, bladder neck contracture (2-6%).
- Retrograde ejaculation 80-90%; ED (erectile dysfunction) 3-5%; incontinence rare, because the sphincter lies outside the plane.

Modern alternatives

- **HoLEP (holmium laser enucleation)** and **RASP (robotic-assisted simple prostatectomy)**: less bleeding and shorter stay; same functional outcome.

Diagram to draw: Sagittal section: capsulotomy and enucleation plane.

Key points to remember:

- Millin 1945: retropubic, transverse capsulotomy ~2 cm below bladder neck, bladder not opened.
 - Apical urethra divided under vision; 5 and 7 o'clock sutures.
 - Freyer (suprapubic) preferred for median lobe, stones, diverticulum.
 - Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.
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II-3d. Energy sources in laparoscopic urology

Type: SAQ (10 marks)

Introduction

- Energy devices cut, coagulate and seal vessels by denaturing protein.
- Electrosurgery uses ~500 kHz radiofrequency current, too fast to stimulate nerve or muscle.
- **Cut:** continuous low-voltage waveform; cell water boils and cells burst. **Coagulation:** interrupted high-voltage waveform; tissue desiccates and heat spreads deeper.

Types of energy sources

Device	Principle	Lateral spread	Notes
Monopolar	Active tip → patient → return pad	5-7 mm	Versatile; stray-current risk
Bipolar	Current only between the jaws	1-6 mm	No pad; no capacitive coupling
Advanced bipolar (LigaSure)	Compression plus impedance-controlled energy fuses collagen	1-3 mm	Seals vessels ≤ 7 mm; single application
Ultrasonic (Harmonic)	Blade vibrating at 25-55 kHz; no current through patient	≈ 1 mm	Safe with pacemakers; blade exceeds 200 °C and cools slowly
Argon beam	Monopolar current through argon gas, non-contact	Superficial	Raw-surface haemostasis; gas embolism

Mechanisms of electrosurgical injury

- Unseen injuries present late as bowel perforation. 1. **Insulation failure:** current leaks through damaged insulation. 2. **Direct coupling:** the active electrode touches another metal instrument. 3. **Capacitive coupling:** current induced through intact insulation, worse with coagulation current; avoid hybrid trocars (metal cannula, plastic anchor). 4. **Thermal spread** to ureter, nerves or bowel. 5. **Return pad burns** when contact area shrinks.

Safety measures

- **AEM (active electrode monitoring)** detects insulation failure and capacitive coupling.
- **REM (return electrode monitoring)** detects poor pad contact.
- Lowest effective power (cut 50-80 W, coagulation 30-50 W), whole tip in view, bipolar near critical structures.

Recent advances

- Smart generators with impedance feedback; robotic vessel sealers.

Diagram to draw: Monopolar vs bipolar circuit; capacitive coupling.

Key points to remember:

- Thermal spread: monopolar 5-7 mm > bipolar 1-6 > LigaSure 1-3 > ultrasonic ≈ 1 mm.
- LigaSure seals vessels ≤ 7 mm; Harmonic vibrates at 25-55 kHz.
- Stray current: insulation failure, direct coupling, capacitive coupling.
- Avoid hybrid trocars; use AEM and REM.

II-3e. Role of vascular grafts in urology

Type: SAQ (10 marks)

Introduction

- Grafts and patches replace, bypass, extend or widen vessels when primary repair is impossible or would narrow them.

Types of grafts

Type	Examples	Advantages	Disadvantages
Autologous vein	Saphenous, gonadal	Resists infection; suits small vessels	Limited length
Autologous artery	Internal iliac, Carrel aortic patch	Ideal for renal arteries	Limited availability
Biological patch	Bovine pericardium	Pliable	Cost
Synthetic	PTFE (polytetrafluoroethylene) , Dacron	Off-the-shelf, large calibre	Infection; venous grafts thrombose

Applications in urology

1. Renal transplantation

- Multiple arteries (18–43%): Carrel patch or bench joining (side-to-side, end-to-side). Preserve the lower pole artery, which supplies the ureter.
- Short right renal vein: extend with donor IVC (inferior vena cava), external iliac or spiral gonadal vein.

2. RCC (renal cell carcinoma) with IVC thrombus: usually cavotomy and primary closure; **patch cavoplasty** if residual lumen would be <50%; **PTFE replacement (16–20 mm)** if the wall is invaded circumferentially.

3. **Post-chemotherapy RPLND (retroperitoneal lymph node dissection)**: vascular procedures in 9–15%.

- IVC: primary repair if narrowing <25%, patch or graft if more; ligation tolerated when collaterals exist.
- Aorta: about 1% need PTFE or Dacron replacement; it can never be ligated.

4. **Renovascular surgery**: aortorenal bypass with saphenous vein or internal iliac artery.

5. **Dialysis access**: PTFE AV (arteriovenous) graft.

Principles

- Autologous grafts for small vessels and contaminated or urine-exposed fields; synthetic for large high-flow vessels.
- Tension-free anastomoses; heparin before clamping.
- Wrap synthetic grafts in omentum to separate them from bowel and urinary tract.

Complications

- Thrombosis (caval grafts 10–40%), stenosis, graft infection (often needing removal), graft-enteric or urinary fistula.

Diagram to draw: Bench reconstruction of two renal arteries.

Key points to remember:

- Autologous for small vessels/contaminated fields; PTFE/Dacron for IVC/aorta.
 - Transplant: Carrel patch, bench reconstruction, IVC/gonadal extension of short right vein.
 - IVC: patch if lumen <50%; PTFE replacement if wall invaded.
 - PC-RPLND: IVC can be ligated, aorta must be grafted.
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II-3f. Transrectal ultrasonography

Type: SAQ (10 marks)

Introduction / Physics

- Piezoelectric crystals convert electricity to sound and back; higher frequency gives better resolution, less penetration.
- Probes are **6-10 MHz** (commonly 7.5 MHz): end-fire, side-fire or biplane.

Technique

- Left lateral position; DRE (digital rectal examination) first; transverse and sagittal scans.

Sonographic anatomy

- McNeal zones are not seen separately. The posterior **peripheral zone** (where ~70% of cancers arise) is homogeneous and echogenic; the **transition zone** (BPH (benign prostatic hyperplasia)) is heterogeneous.
- Corpora amylacea outline the surgical capsule. Seminal vesicles are symmetrical, about $4.5\text{--}5.5 \times 2$ cm.

Measurements

- **Volume (ellipsoid) = $0.52 \times \text{transverse} \times \text{AP (anteroposterior)} \times \text{length}$** ; 1 mL $\approx$ 1 g; normal adult gland 20–25 g.
- **PSAD (PSA (prostate-specific antigen) density) = PSA/volume**; >0.15 is suspicious.

Findings

- **Cancer:** classically a hypoechoic peripheral-zone lesion, but 39% are isoechoic. False positives: BPH nodules, prostatitis.
- **Cysts:** midline Müllerian cyst; off-midline ejaculatory duct cyst (azoospermia).

Indications

- Biopsy guidance; volume before BPH surgery or brachytherapy.
- Azoospermia work-up (ejaculatory duct obstruction); prostatic abscess drainage.

TRUS (transrectal ultrasound)-guided biopsy

- Systematic **12-core** biopsy under periprostatic lidocaine block.
- Contraindications: coagulopathy, acute prostatitis, severe immunosuppression.
- Complications: haematuria, haemospermia, rectal bleeding, and **sepsis** (0.3–3%) from FQ (fluoroquinolone)-resistant E. coli.
- Prophylaxis: targeted (rectal swab) or augmented; fluoroquinolones are now restricted.

Current shift

- **EAU (European Association of Urology)** recommends transperineal biopsy because of lower infection risk.
- **Pre-biopsy mpMRI (multiparametric MRI)** with targeted biopsy:
 - **PROMIS (2017):** mpMRI sensitivity for significant cancer 93% versus 48% for TRUS biopsy.
 - **PRECISION (2018):** MRI-targeted biopsy detected more significant cancer (38% vs 26%).

Recent advances

- **Micro-ultrasound (29 MHz):** near-histological resolution; comparable to mpMRI for targeting.

Diagram to draw: Transverse and sagittal views; 12-core biopsy template.

Key points to remember:

- Probe 6–10 MHz; volume = $0.52 \times T \times AP \times L$; PSAD >0.15 suspicious.
 - 39% of cancers are isoechoic; 12-core systematic biopsy with periprostatic block.
 - Sepsis 0.3–3%; EAU prefers transperineal biopsy.
 - PROMIS and PRECISION established the MRI-first pathway.
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Rapid Revision Sheet - Paper II

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- IPSS 0–35: 0–7 mild, 8–19 moderate, 20–35 severe.
- 5-ARIs shrink the prostate ~20%, halve PSA; PLESS cut AUR/surgery 51%; MTOPS combination cut progression 66%.
- AUA sizes: TUIP ≤ 30 ; TURP/PVP/PUL/Rezūm/Aquablation 30–80; HoLEP any size; simple prostatectomy > 80 mL.
- TUR syndrome: gland > 45 mL, resection > 90 min; bipolar TURP avoids it.

II-2. Discuss metabolic evaluation in urinary calculus disease. (LAQ)

- 24-h urine on usual diet, ≥ 4 weeks after event; check creatinine for adequacy.
- Targets: volume > 2.5 L, oxalate < 40 mg, citrate > 320 mg, Ca $< 250/300$ mg (or < 4 mg/kg).
- Distal RTA: NH_4Cl load fails to lower pH < 5.5 ; cystinuria: cystine < 250 mg/L, pH 7.0–7.5, tiopronin.
- Borghi 1996: fluid halves recurrence; repeat 24-h urine within 6 months.

II-3a. Management of priapism (SAQ)

- Ischaemic: $\text{pO}_2 < 30$, $\text{pCO}_2 > 60$, pH < 7.25 – emergency.
- Phenylephrine 100–500 $\mu\text{g/mL}$, 1 mL q3–5 min up to 1 h.
- Distal shunt $\pm$ tunnelling before proximal; prosthesis if > 36 h.
- High-flow: observe ($\sim 60\%$ resolve), then embolise.

II-3b. Renal tubular acidosis (SAQ)

- Type 1: urine pH > 6 , hypokalaemia, hypocitraturia, calcium phosphate stones (up to 70%).
- Type 4: hyperkalaemia, low urine pH; seen in obstructive uropathy.
- Positive UAG = renal cause; NH_4Cl load – pH fails to fall < 5.5 in distal RTA.

II-3c. Millin's prostatectomy (SAQ)

- Millin 1945: retropubic, transverse capsulotomy ~ 2 cm below bladder neck, bladder not opened.
- Apical urethra divided under vision; 5 and 7 o'clock sutures.
- Freyer (suprapubic) preferred for median lobe, stones, diverticulum.
- Retrograde ejaculation 80–90%; transfusion $< 5\%$; bladder neck contracture 2–6%.

II-3d. Energy sources in laparoscopic urology (SAQ)

- Thermal spread: monopolar 5–7 mm $>$ bipolar 1–6 $>$ LigaSure 1–3 $>$ ultrasonic ≈ 1 mm.
- LigaSure seals vessels ≤ 7 mm; Harmonic vibrates at 25–55 kHz.
- Stray current: insulation failure, direct coupling, capacitive coupling.
- Avoid hybrid trocars; use AEM and REM.

II-3e. Role of vascular grafts in urology (SAQ)

- Autologous for small vessels/contaminated fields; PTFE/Dacron for IVC/aorta.
- Transplant: Carrel patch, bench reconstruction, IVC/gonadal extension of short right vein.
- IVC: patch if lumen $< 50\%$; PTFE replacement if wall invaded.
- PC-RPLND: IVC can be ligated, aorta must be grafted.

II-3f. Transrectal ultrasonography (SAQ)

- Probe 6–10 MHz; volume = $0.52 \times T \times AP \times L$; PSAD > 0.15 suspicious.

- 39% of cancers are isoechoic; 12-core systematic biopsy with periprostatic block.
- Sepsis 0.3–3%; EAU prefers transperineal biopsy.
- PROMIS and PRECISION established the MRI-first pathway.